Day 4: Polymer Chain MD Simulation with GROMACS

In this Day 4 workshop, we performed a complete molecular dynamics (MD) simulation workflow for a polymer chain in water using **GROMACS**. The goal was to take a polymer structure generated in previous sessions and prepare it for realistic MD simulations, following best practices for system setup, minimization, equilibration, and basic analysis.

Table of Contents

Initialization

The first step is to set up the environment for the tutorial. This includes:

- Opening the terminal and navigating to the `polymermd-workshop-main` folder:

  ```
  cd /PATH_TO/polymermd-workshop-main
  ```

  Replace *PATH_TO* with the directory path on your computer.

- Activating the conda environment:

  ```
  conda activate polymer-md
  ```

- Launching Jupyter Lab:

  ```
  jupyter lab --ip 0.0.0.0 --no-browser
  ```

Open the printed URL in your browser, or configure it to open automatically.

Key Steps and Learning Objectives

- **System Preparation:**
  - Copied the polymer structure (`.gro`) and topology (`.top`) files from the Day 3 output.
  - (Optionally) Performed a short pre-equilibration to relax the polymer and reduce the box size, ensuring a more compact and physically reasonable starting structure.
- **Solvation and Ion Addition:**
  - Added a cubic water box around the polymer and solvated the system with SPC/E water molecules, mimicking a realistic aqueous environment.
  - Added Na+ and Cl- ions to neutralize the system and achieve a target ionic strength, which is essential for simulating biological or soft-matter systems.
- **Energy Minimization:**
  - Performed energy minimization to remove steric clashes and relax the initial structure, ensuring the system is stable before running dynamics.
- **Equilibration:**
  - Ran NVT (constant Number, Volume, Temperature) equilibration to stabilize the temperature of the system.
  - Followed with NPT (constant Number, Pressure, Temperature) equilibration to stabilize the pressure and density, allowing the box size to adjust and the system to reach equilibrium conditions.
- **Analysis and Extraction of Properties:**
  - Used `gmx energy` to extract key thermodynamic properties (temperature, volume, density) from the simulation output.
  - Visualized these properties to assess the quality of equilibration and system stability.
- **Visualization:**
  - Used Python tools such as **nglview** and **MDAnalysis** to visualize the polymer structure and trajectories, providing insight into the system's behavior at each stage.

Throughout the workshop, we emphasized the importance of careful system setup, stepwise equilibration, and validation of simulation results. Each step was accompanied by practical GROMACS commands and Python code for visualization and analysis, giving participants hands-on experience with a typical MD workflow for polymers in solution.

GROMACS Input Formats and Commands

In this tutorial, we used several standard GROMACS file formats and core commands:

Key Input/Output File Formats:

- `.gro`: GROMACS coordinate file containing atomic positions and box dimensions.
- `.top`: Topology file describing the molecular structure, force field parameters, and system composition.
- `.mdp`: Molecular dynamics parameter file specifying simulation settings.
- `.tpr`: Portable binary run input file generated by `gmx grompp` (contains all info for a simulation run).
- `.edr`: Energy file containing simulation energy data.
- `.trr`/`.xtc`: Trajectory files (full-precision and compressed, respectively).
- `.log`: Log file with simulation details.

Main GROMACS Commands Used:

- `gmx grompp`: Preprocesses input files and creates the `.tpr` run file.
- `gmx mdrun`: Runs the actual MD simulation or energy minimization.
- `gmx editconf`: Defines the simulation box and manipulates coordinates.
- `gmx solvate`: Adds solvent (e.g., water) to the simulation box.
- `gmx genion`: Adds ions to the system for neutralization and ionic strength.
- `gmx energy`: Extracts energy and thermodynamic properties from simulation output.

These commands form the backbone of a typical GROMACS workflow, from system setup to analysis.

Simulation Preparation Steps

1. Copy Initial Polymer Files

Copy the polymer structure (`.gro`) and topology (`.top`) files generated in Day 3 to your working directory. For example, with `polymer_name = "PE025_PET25_random"`:

Run in: `simulations/PE025_PET25_random/`

```
cp ../../../Day3-gen-polymer-solved/polymers/PE025_PET25_random/polymer*gro polymer.gro
cp ../../../Day3-gen-polymer-solved/polymers/PE025_PET25_random/topol.top topol.top
```

2. Pre-Equilibration (Optional)

Relax the single polymer to reduce the simulation box size.

Run in: `simulations/PE025_PET25_random/`

```
gmx grompp -f preeq.mdp -c polymer.gro -p topol.top -o preeq.tpr -maxwarn 2
gmx mdrun -s preeq.tpr -deffnm polymer_relax -v
```

3. Solvate the Polymer

Add a cubic water box and solvate the system with SPC/E water molecules.

Run in: `simulations/PE025_PET25_random/`

Note: After solvation and before adding ions, you may need to manually edit `topol.top` to:

- Add or include water and ion parameters (e.g., `#include "amber03.ff/tip3p.itp"` for water, and ion definitions for Na+ and Cl-).
- Update the number of solvent and ion molecules as reported by GROMACS after solvation and ion addition.

```
gmx editconf -f polymer_relax.gro -o polymer_box.gro -d 1.0 -bt cubic
gmx solvate -cp polymer_box.gro -cs spc216.gro -o polymer_solv.gro -p
topol.top -maxsol 1000
```

4. Add Ions

Neutralize the system and add Na+ and Cl- ions to achieve the desired concentration.

Run in: `simulations/PE025_PET25_random/`

```
gmx grompp -f minimization.mdp -c polymer_solv.gro -p topol.top -o
ions.tpr
echo SOL | gmx genion -s ions.tpr -o polymer_solv_ions.gro -p topol.top -
pname NA -nname CL -neutral -conc 0.01
```

Energy Minimization

Relax the system to remove steric clashes and optimize geometry.

Run in: `simulations/PE025_PET25_random/em/`

```
gmx grompp -f em.mdp -c ../polymer_solv_ions.gro -p ../topol.top -o em.tpr
-maxwarn 2
gmx mdrun -s em.tpr -deffnm em -v
```

Equilibration

1. NVT Equilibration

Stabilize the system at constant number of particles (N), volume (V), and temperature (T).

Run in: `simulations/PE025_PET25_random/nvt/`

```
gmx grompp -f nvt.mdp -c ../em/em.gro -p ../topol.top -o nvt.tpr -maxwarn 2
gmx mdrun -s nvt.tpr -deffnm nvt -v
```

2. NPT Equilibration

Stabilize the system at constant pressure (1 bar) and temperature (300 K).

Run in: `simulations/PE025_PET25_random/npt/`

```
gmx grompp -f npt.mdp -c ../nvt/nvt.gro -p ../topol.top -o npt.tpr -maxwarn 2
gmx mdrun -s npt.tpr -deffnm npt -v
```

Extracting Energy Terms

After equilibration, you can extract properties such as temperature, volume, and density from the energy file using `gmx energy`.

Run in: `simulations/PE025_PET25_random/npt/`

```
gmx energy -f npt.edr -o energy.xvg
```

This will prompt you to select energy terms. For example, select:

- 14: Temperature
- 20: Volume
- 21: Density

Press enter again to finish selection and generate the output file.

Example snapshot:

energy-term-selection

The resulting `energy.xvg` file can be plotted in Python to visualize temperature and density over time.

Example output:

energy-xvg-output

Visualization

Visualize the polymer structure or trajectory using **nglview** or **MDAnalysis** in Python:

```
visualize_trajectory("polymer_solv_ions.gro")
```

License